YEDITEPE UNIVERSITY
FACULTY OF ENGINEERING

COURSE SYLLABUS
2026/2027-1

Course Code-Name	**CSE232 Systems Programming**

Course Schedule	Wed 13:00-13:50, Thu 11:00-12:50, Lab. Thu 16:00-17:50
Instructor's Name **Phone** **E-mail** **Office Hours**	Esin Onbaşıoğlu (216) 578 04 22 (x-1422) esin@cse.yeditepe.edu.tr Thu 13:00-14:50
Assistant's Name **Phone** **E-mail**	Sarper Özer sozer@ cse.yeditepe.edu.tr
Textbook & Supplementary Materials	Course Notes: on YULEARN W. Wray, J. Greenfield, R. Bannatyne, Using Microprocessors and Microcomputers, Prentice-Hall L.A. Leventhal, 6800 Assembly Language Programming, Osborne & Associates E. Borin, An Introduction to Assembly Programming with RISC-V, online L.L. Beck, System Software, Addison-Wesley D.H. Marcellus, Systems Programming for Small Computers, Prentice Hall J.J. Donovan, Systems Programming, McGraw Hill A. Silberschatz, et al., Operating System Concepts, Addison-Wesley
Prerequisite	CSE 114
Course Outline	Week-1 Introduction: Numbering system, hardware, I/O, storage, systems software, assembly language Week-2,3 Assembly Language Programming: M6800 Instruction set, conditional instructions, loops, subroutines, time delays, basic I/O Week-4 Assembly Language Programming: RISC V Instruction set, conditional instructions, loops, subroutines Week-5 Assemblers: Two-pass assemblers, relocation Week-6 Linkers and Loaders: subroutine linkage, library linkage, absolute/relocating/direct-linking loaders, dynamic linking Week-7 Linux Week-8 Macros and macro processors: One-pass and two-pass designs Week-9 Editors: Text editors, file editing, insertion, deletion, etc. Week-9 Debuggers Week10 Formal specification of programming languages and introduction to compilers Week-11,12 Interpreters: statement parsing, symbol table, processing of statements Week-13,14 Introduction to operating systems: user interface, run-time environment, interrupts, machine-dependent and machine-independent features
Midterm Date	19.11.2026
Makeup lectures	28.11.2026, 5.12.2026
Grading	Midterm 25% Final 40% Laboratory 15% Term Project 10% Quiz 10% (quiz dates will not be announced)

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Attendance	▪ **You must attend at least 80% of the lectures and 80% of the lab hours. Otherwise you will get FA.**
Course Objectives	Knowledge in the design and implementation of system programs i.e. assemblers, linkers, loaders, macro-processors, editors, interpreters, compilers and operating systems; assembly language programming; UNIX shell programming
Additional Remarks	

Course Outcomes

i. Adequate knowledge in mathematics, science and engineering subjects pertaining to the relevant discipline; ability to use theoretical and applied information in these areas in complex engineering problems.

iv. Ability to devise, select, and use modern techniques and tools needed for analyzing and solving complex problems encountered in engineering practice; ability to employ information technologies effectively.

v. Ability to design and conduct experiments, gather data, analyze and interpret results for investigating complex engineering problems or discipline specific research questions.

vi. Ability to work efficiently in intra-disciplinary and multi-disciplinary teams; ability to work individually.